5

Optimality Theory and Syntax: Movement and Pronunciation*

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The notion of "optimization" standardly assumed by Optimality Theory is quite simple and natural. While not uncontroversial, it has advanced our understanding of many long–standing problems in phonology. Nonetheless, it is not the *only* simple and natural proposal to enter linguistic theory. The successes of OT within phonology and morphology) do not *necessarily* teach us anything more general about how other aspects of language work. On the other hand, they might.

It would be interesting, for example, if we were to discover that the mechanics of constraint interaction proposed by Prince and Smolensky (1993) characterize *all* aspects of linguistic knowledge and use. Such a discovery would reveal a striking uniformity across linguistic processes that would immediately raise wider questions about possible uniformity of interaction beyond the boundaries of language. Is the human language faculty, or the mind, more broadly, an optimizer?

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*A short, more technical survey of the work discussed here appears in Pesetsky (to appear), but the work is part of a much larger endeavor. The presentation in this chapter is intended to illustrate some results and issues, and does not at every point present its material with the care that might constitute proof. Syntactically informed readers will recognize many cut corners and roads not taken. I am grateful for discussion of this work to audiences at the University of Arizona, Eotvos Lorand University, the Generative Grammar Circle of Korea, the Holland Institute of Generative Linguistics, the Numazu Summer Seminar in Linguistics, Sophia University, DIPSCO at Ospitale San Raffaele, the University of Paris-Nanterre, the University of Pennsylvania, UCLA and USC, as well as to students at MIT. Particularly helpful were discussions with Noam Chomsky, Danny Fox, Paul Hagstrom, Richard Kayne, Martha McGinnis, Eric Reuland, Donca Steriade, Hubert Truckenbrodt and Ken Wexler. Tom Bever provided useful comments on the lecture from which this chapter was developed.

It would be equally interesting, however, if we were to learn something entirely different: that optimality–theoretic interactions characterize *some, but not all*, aspects of language. We would then face the intriguing task of discovering exactly what Optimality Theory is and is not good for — and, ultimately, understanding the reasons for any divisions of labor that we may find.

Clearly, these are issues that can only be addressed through serious investigation. These are not questions that can be given useful *a priori* answers. The initial case for OT in phonology rested on conceptual flaws in standard accounts of phenomena like Berber syllabification and the cross–linguistic character of infixation. These phenomena display such an obviously optimality–theoretic character that alternative, more traditional treatments seemed to miss the point. In parallel fashion, we should ask whether any problems of syntax have a similar character. Are there problems connected with the grammar of sentences for which traditional, non–OT treatments patently miss the point somehow. If the answer is "yes", we must also ask whether there are areas in which it is the *OT* treatment that "misses the point". Once we have a body of work that asks and answers both types of questions, we will be on our way towards a general understanding of the scope and nature of optimization phenomena in human language.

This chapter and the next sketch what such an investigation might look like in the domain of **syntax.** In order to pursue this goal, however, we first need a general understanding of what is known about syntax. I therefore begin with a sketch of the "state of the art", enumerating some discoveries about sentence structure that seem the most profound and important. Only once we gain some clarity of vision about the field will we be in a position to ask where (if at all) ideas from Optimality Theory are relevant.

I will suggest that there is only weak evidence for OT constraint interaction in several areas traditionally viewed as central to syntax, and in fact some serious arguments against an OT view in these domains. On the other hand, other areas of syntax — in particular those that lie at the boundary between syntax and phonology — have a strikingly obvious OT character. This is no surprise, if OT interactions are a hallmark of aspects of language closest to the mouth and ear, but is perhaps a surprise if OT is a "theory of everything". The technical discussion in this chapter will therefore bear directly on the fundamental questions with which the chapter began.

We need to start with a few words of warning. The discussion in this chapter is not only pedagogical (cutting a number of corners), but also speculative and tentative. The investigation of OT properties of syntax is in its infancy. Evidence for OT interactions might be strongest at the boundaries with phonology merely because we know more about OT in the domain of phonology — or it may be the case that OT really *is* a theory of phonology. This chapter will not decide these matters, but will try to point out how they might eventually be decided.

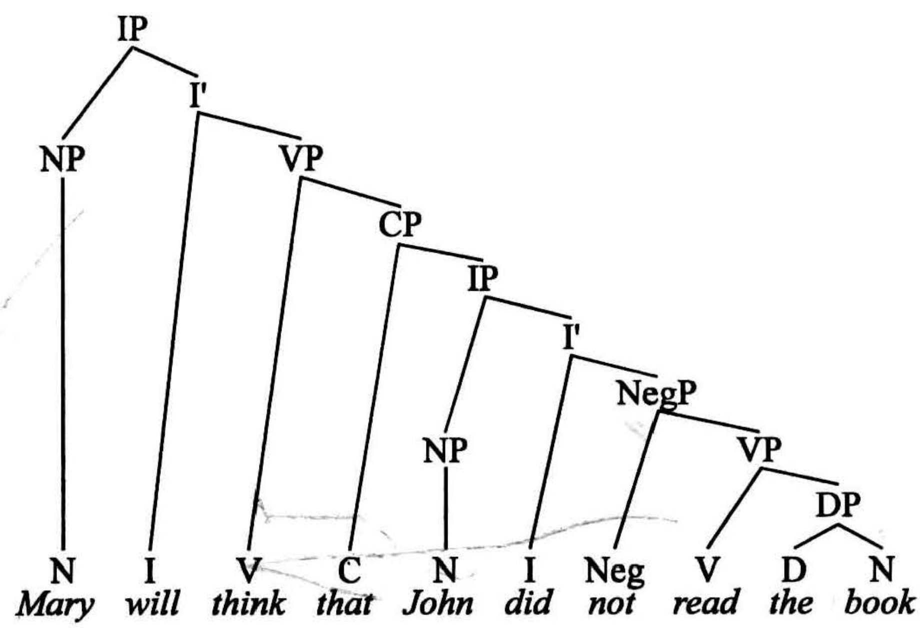
1 The Components of Syntax: Constituency and Movement

The history of syntactic investigation is marked by a small number of central discoveries which created the syntactician's research agenda. One can divide these discoveries into two groups — the discovery of **hierarchical constituent structure**, and the discovery that elements may occupy more than one position within this hierarchy, which the literature calls **movement**, for reasons we shall shortly discuss.

From Grammatical Relations to Constituent Structure

One of the most important syntactic achievements, traceable to the work of pre–modern grammarians, is the discovery of **grammatical relations**. This discovery involves, first, the realization that words group together to form phrases, and second, that not all phrases are created equal. Thus for example, there is a difference between phrases that function as subject–of–the–sentence and phrases that function as object–of–the–sentence. The discovery was understandably abetted by the encoding of such distinctions in the morphology of languages like Greek, Latin, Sanskrit, Japanese — but crucially involved the recognition that an *abstract* notion lay behind superficial facts of morphology.

(5.1) Hierarchical constituent structure



Legend:
C = complementizer (subordinating conjunction)
D = determiner (article, pronoun, possessive *'s*)[1]
I = inflection
N = noun
Neg = negation
V = verb
XP indicates maximal phrase headed by X (VP = verb phrase, etc.)
X' indicates intermediate phrase headed by X (read as "X–bar")

[1]In the structures drawn in this chapter, I do not assume that D and DP are present when there is no determiner word present. That is why *Mary* is an NP, but *the book* is labeled DP. This assumption might not in the end be correct, but will suffice for our purposes.

In this way, for example, languages that lack rich morphology could be insightfully described using the same categories that make obvious sense in languages with rich morphology, and a general understanding of human–language syntax could be developed.

A related discovery was the realization that the phrasal organization of sentences is **hierarchical** — generally characterizable by the sort of **phrase–structure** tree exemplified in (5.1) for the English sentence *Mary will think that John did not read the book.* Thus, while words *the book* are a noun phrase (NP) in (5.1), they also form part of a larger phrase that properly contains it — the verb phrase (VP) *read the book.* That verb phrase, in turn, forms part of larger phrases *not read the book, did not read the book,* etc. — each of which in turn may be composed of a set of phrases. One element of each phrase, called its **head,** determines its major properties. Phrases are **labeled** to reflect the identity of their heads; see the Legend to example (5.1) above.

Constituency reveals itself in many ways. For example, the constituents of a sentence are the fragments that can be uttered in surprise.[2] So, if I never knew the fact communicated by sentence (5.1) before, I might exclaim in surprise (5.2), repeating the constituent labeled **I'**. But I can't utter in surprise (5.3), picking out a nonconstituent.[3]

(5.2) Did not read the book?? You must be kidding!

(5.3) *Did not read the?? You must be kidding!

Syntactic constituents bear a law–like relation to semantic chunks (though the correspondence is often more complex than one–to–one). For example, the fact that the proposition *that John did not read the book* functions as an argument of the verb *think* corresponds to the fact that this sequence forms a syntactic constituent.

In addition, the verb of the sentence combines with the direct object to form a constituent that excludes the subject. Consequently, verb+object, but not verb+subject, can be a sentence fragment uttered in surprise.

(5.4) Sue unwrapped the present.
a. Unwrapped the present?? You must be kidding!
b. *Sue unwrapped?? You must be kidding!

This fact about verbs and direct objects — a possible universal fact about languages — will be important in the next section.

Constituent structure also places limits on the ways in which phrases may **corefer.** In (5.5a), we cannot understand the pronoun as referring to the same individual as *John.* (That is, *John* cannot function as the **antecedent** for the pronoun.) On the other hand,

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[2]I owe the "sentence fragment test" for constituency to Radford (1988), a standard textbook covering some of the fundamentals discussed in this chapter. A more comprehensive survey is provided by other textbooks, especially Haegeman (1994).

[3]The fragment *I don't remember the* naturally occurs when the speaker can't think of the word that follows *the*. This is not a counterexample because the speaker intended to utter a constituent.

(5.5b) is different. Here the pronoun *his* may refer to *John*. (*John* may function as the antecedent for *his*.)

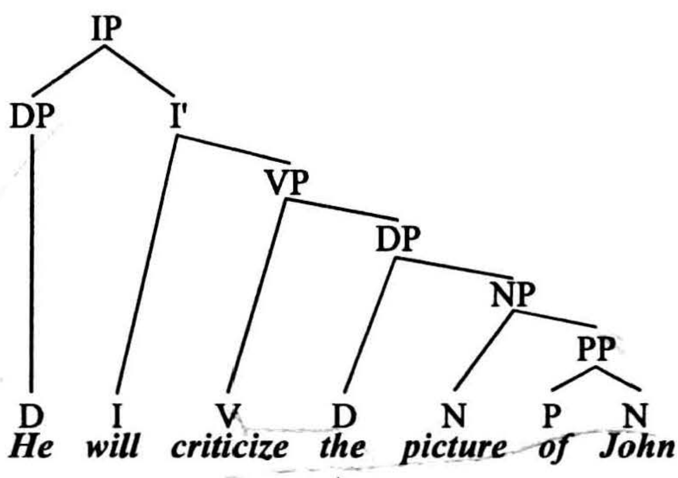
(5.5) Patterns of coreference and non–coreference

a. He will criticize the picture of John.
[*He* cannot be *John*]

b. His former teacher will criticize the picture of John.
[*His* may be *John*]

The source of this difference does not lie in the fact that the pronoun precedes its would–be antecedent, since this is true of both examples. Instead, the source of the difference is structural, namely that the pronoun *he* in (5.5a) is attached to the tree at a node that contains its antecedent *John*. That is, the first available **sister** node to *he* is the phrase *will criticize the picture of John*, and that phrase contains *John*. Syntacticians call this relation **c–command**. We say that *he* c–commands *John* in (5.5a).

(5.6) Structure of (5.5a)

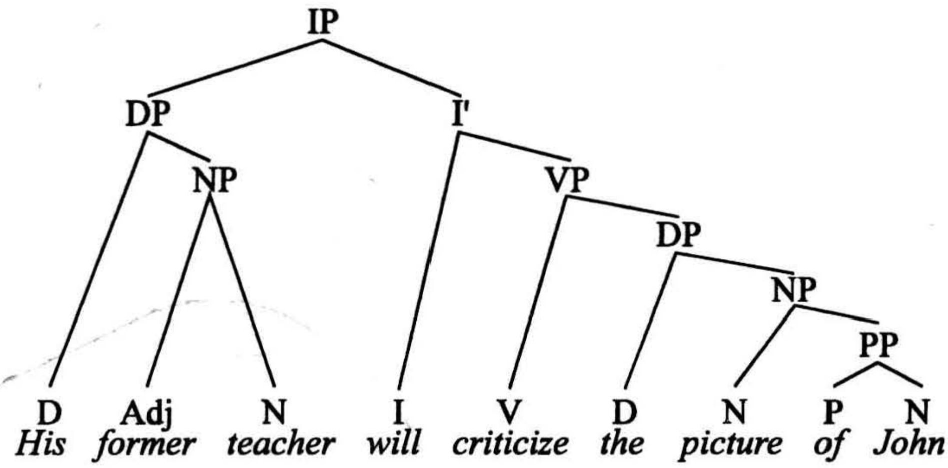


Legend:
P = preposition

By contrast, the first sister node available to the determiner *his* in (5.5b) is the NP *former teacher*, which does not contain *John*. Consequently, *his* in (5.5b) does not c–command *John*. This means that *John* can function as the antecedent for *his*.[4] Examining many contrasts of this sort, linguists have generally concluded that coreference between a pronoun and a non–pronoun is restricted by c–command relations of just this sort. A pronoun may precede its antecedent, but: may not c–command it.

[4]Note also that ours is a fact about *structure*, not about the difference between *he* vs. *his*. For example, in *his picture of John*, the pronoun *his* cannot corefer with *John* because it c–commands it, even though *his* and *John* may corefer in (5.5b).

(5.7) Structure of (5.5b)



> The fact that so many phenomena refer to constituency — the grammar of sentence fragments, semantic interpretation, limitations on coreference, and much more besides — serves as evidence that constituent structure is a central property of syntax.

Movement

The second central property of syntax that we examine is the discovery that some phrases and words bear *more than one* grammatical relation, i.e. that *some phrases and words occupy more than one structural position* in the hierarchical structures just discussed.

Consider, for example the rules that govern the placement of the verb in standard German. Every foreign student of German has run afoul of the fact that the position of the German verb depends on the type of clause it occupies. A common description of the situation is the pair of statements in (5.9).

(5.9) German verb placement: a common description

a. In a main clause, the tensed verb follows the first phrase.
["verb–second"]

b. In a subordinate clause, the tensed verb comes last.
["verb–final"]

These statements accurately describe the differences the placement of **sah** in (5.10a–b).

(5.10) **a.** Gestern **sah** Hans den Mann.
yesterday saw Hans the man
'Yesterday, Hans saw the man.'
[main clause, verb–second]

b. Marie denkt, dass Hans gestern den Mann **sah**.
Mary thinks, that Hans yesterday the man saw.
'Mary thinks that yesterday Hans saw the man.'
[subordinate clause, verb–final]

From a cross–linguistic perspective, (5.10a) is a puzzle, since the verb is separated from its object by the subject of the sentence. Although the point is somewhat controversial, it is possible that the verb and its object universally form a constituent that excludes the subject, just as they do in the English structures seen above.[5] This is obviously not possible when the verb and object are separated by the subject.

In fact, there are cases in which the subject separates the verb not only from its object — but also from a piece of itself. In these cases, most of the verb occupies second position, and another piece (the so–called **separable prefix**) is pronounced in a different position. Interestingly, the prefix is pronounced at the end of the VP — right next to the direct object. The separable prefix is, in fact, located exactly where the verb as a whole would sit in a subordinate clause. To make this concrete, let us examine some sentences whose main verb is *anmachen* 'turn on'. This verb consists of a morpheme *an–* (which otherwise means 'on') and a root *machen* ('make'). In a subordinate clause, *anmachen* shows up in one piece in V–final position. In a main clause, however, only the morpheme *machen* shows up in second position. The morpheme *an* shows up in final position, i.e. the spot where the whole verb is pronounced in subordinate clauses:[6]

(5.11) **a.** Wir –machen jetzt das Licht an–.
we make now the light on
'We are now turning on the light.'
[main clause, verb–second]

b. Marie denkt, dass wir jetzt das Licht anmachen.
Mary thinks that we now the light on–make
'Mary thinks that we are now turning on the light.'
[embedded clause, verb–final]

It is as if the verb *anmachen* had its pronunciation spread out over two separate structural positions — the "normal" verb position at the end of the sentence next to the direct object, and the verb–second position near the front of the sentence. In this way, we see a situation in which a single word *anmachen* seems to occupy two positions at once — with pieces of the word pronounced in each position.

In fact, we can say more about these two positions. The verb–final position of subordinate clauses is the expected position for the verb in a language where the heads of phrases are final in their phrase. There are many languages in which the verb is final within the verb phrase — for example, Japanese (see Chapter 6).

(5.12) Hanako–wa susi–o tabeta.
Hanako–TOPIC sushi–OBJ ate
'Hanako ate sushi'

[5]Some recent work develops this idea in a manner different from that seen in (5.1). I have presented the older, structurally simpler theory for expository reasons.

[6]The significance of these cases was first noticed in the early 1960s by Bach (1962) and by Bierwisch (1963).

The nature of the verb–second position is less obvious and more interesting. Den Besten (1983) was perhaps the first to point out the significance of some important classes of exceptions to the standard description of German word order in (5.9) — certain subordinate clauses with verb–second order. For example, in addition to (5.13a) with the expected verb–final order in the subordinate clause, (5.13b) is also possible. Example (5.13b) shows verb–second word order in the embedded clause, as well as an unexplained change of mood to subjunctive. Likewise, in addition to (5.14a), we also find (5.14b), which once again shows verb–second word order in the embedded clause. The clauses of interest in the following examples are therefore the subordinate clauses, in which the verb of interest is highlighted in boldface.

(5.13) Main clause word order in embedded clauses

a. Hans sagte, dass er glücklich **ist**.
Hans said that he happy is
'Hans said (that) he is happy.'
[verb–final]

b. Hans sagte, er **sei** glücklich.
Hans said he be happy
[verb–second]

(5.14) **a.** Er benahm sich, als ob er noch nichts gegessen **habe**.
he behaved himself as if he yet nothing eaten had
'He behaved as though he had eaten nothing.'
[verb–final]

b. Er benahm sich, als **habe** er noch nichts gegessen.
he behaved himself as had he yet nothing eaten
[verb–second]

What induces verb–second is not main clauses, but clauses that are not introduced by a complementizer. That is why verb–second order shows up when *dass* is missing in (5.13b) and *ob* is missing in (5.14b). Consequently, it is tempting to suggest that the verb in verb–second clauses in fact occupies the complementizer position. The same conclusion can be drawn from (5.13b), with one caveat. The verb is in second, rather than first position within CP, because of the existence of one extra position to the left of the complementizer. This position is often called the "specifier of CP" — or **SPEC(CP)**. The topic of the sentence occupies SPEC(CP), as do certain other elements, including clause–initial adverbs. Many or all phrases seem to have a unique specifier position of this sort. For example, the subject of the sentence is the SPEC(IP). Some people also suggest that the possessor phrase in a nominal (*Bill*'s *book*) is the SPEC(DP) — with the *'s* occupying the D position. Den Besten's hypothesis thus holds that the verb in German *occupies two positions* in certain clauses, and that the position to the left of the verb in verb–second clauses is the SPEC(CP).

In other words, German only *seems* to display a difference between main vs. subordinate clauses. The reason is the interaction of the "better description" in (5.15) with the entirely independent fact that main clauses are not usually introduced by a subordinating conjunction. The grammatical representations for (5.10a–b), then, are (5.16a–b), with the shaded box indicating the complementizer position — containing the complementizer in (5.16a), and the tensed verb in (5.16b). The line through the second verb–final occurrence of *sah* in (5.16a) indicates that the verb occupies this position, but is not pronounced there.

(5.15) **German verb placement: a better description**

a. The verb occupies the rightmost position within the VP.

b. When the complementizer of a clause is not pronounced, the finite verb of the clause not only occupies the rightmost position within the VP, but also occupies the position of the complementizer.

(5.16) **German verb in the complementizer position**

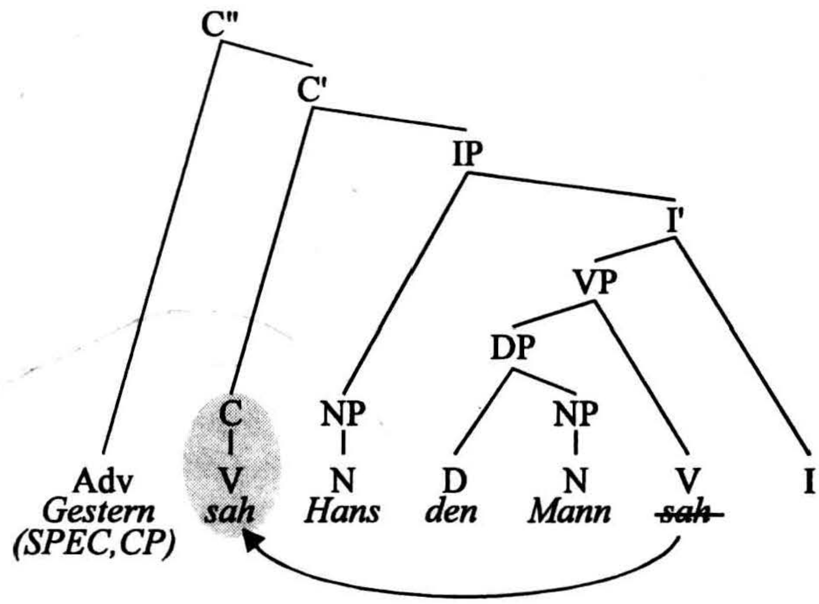
a. Gestern	sah	Hans	den	Mann	~~sah.~~	
Yesterday	saw	Hans	the	man	saw	
b. Marie denkt,	dass	Hans	gestern	den	Mann	**sah.**
Mary thinks	that	Hans	yesterday	the	man	saw

One important note about this phenomenon. For our purposes, all we need to assume is the concept just developed: the idea that certain words and phrases are associated with more than one structural position. Nonetheless, there is some evidence supporting a more elaborate view, according to which the positions that a phrase occupies are arranged in a sequence. According to this view, assignment of a phrase to more than one position is the consequence of a **movement** procedure that copies a word or phrase into the various positions of this sequence.[7] The view of verb–second in German as *movement of the finite verb to the complementizer position* is exemplified in (5.17).[8]

[7]Chomsky's (1995b) **Minimalist Program** (see Chapter 6) adopts the sequential view not only for movement, but also for the laws that govern basic constituent structure. The fundamental objects of syntax in this theory are constructed by the rules **Merge** (form constituent) and **Move**. Nothing in this chapter bears on the correctness of this view, often called "derivational". OT work in phonology generally does not motivate derivational notions as part of constraint evaluation, but does leave open the possibility of a derivational character for GEN, the procedures that yield the candidates that the constraints evaluate.

[8]Some notes on (5.17). First, the adverbial in SPEC(CP) also occupies that position by virtue of movement. Second, the I node here is not filled by an auxiliary verb (like have), but is probably also a position occupied by the main verb. Movement of V to C is actually movement of V to I, and then to C. Finally, in other work (Pesetsky to appear), I argue that the verb in C is actually attached to the left of an unpronounced complementizer, rather than occupying the actual complementizer position as in (5.17). If you are reading this chapter for the second time, you can see for yourself that the discussion developed below explains why the complementizer must be unpronounced. These details can be ignored for present purposes.

(5.17) Movement of V to C



Since the term "movement" is so common, I will adopt it here — but the sequential aspect of this description will play no role in our discussion.[9] What is important to us about movement is the basic discovery that a word or phrase may be associated with more than one position.

Movement phenomena abound in the languages of the world. Furthermore, every now and then, we are lucky enough to find instances of movement in which, as with German prefixed verbs, one can actually hear the pronunciation of a syntactic unit "spread across" more than one syntactic position. Japanese furnishes another telling example. The Japanese verb and its inflections are quite rigidly located at the end of the sentence,[10] but in many other respects, word order is quite free. For instance, the Japanese rendering of *Taroo gave the book to Hanako* may show any order of constituents — so long as the verb remains at the end. This sort of freedom is called **scrambling**.

Armed with our understanding of constituent structure and movement, we might be tempted to attribute this variation in word order to movement. Such a proposal would allow us to analyze Japanese as a language with a rich constituent structure and a simple fundamental word order: Subject – Indirect Object – Direct Object – Verb. Movement of the subject and objects would explain the fact that subject and object sometimes show up in unexpected positions. But then remorse and guilt might set in. "How Anglocentric of us to impose on Japanese the rigidities of English word order," we might think.

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[9]Some schools of thought in modern syntax assume a view of "movement" phenomena in which significantly less information may be shared among the various positions than in the model discussed in this chapter. For example, the research tradition called first **Generalized Phrase Structure Grammar** (GPSG; see Sells 1985; Gazdar, Klein, Pullum & Sag 1985) and later **Head–Driven Phrase Structure Grammar** (HPSG; Pollard & Sag 1994) differs from our proposal in this manner. The reader is invited to consider how these alternative traditions might handle the evidence presented in this chapter in favor of full multiple association (movement), but I will not undertake this sort of comparison here.

[10]There are arguments that the Japanese verb moves from VP–final position rightwards to a CP–final position (Koizumi 1995). These arguments do not affect our discussion.

(5.18) **Scrambling in Japanese**

a. Taroo–ga Hanako–ni hon–o ageta.
Taroo–SUB Hanako–to book–OBJ gave
'Taroo gave the book to Hanako.'

b. Hon–o Taroo–ga Hanako–ni ageta.
book–OBJ Taroo–SUB Hanako–to gave

c. Hanako–ni Taroo–ga hon–o ageta.
Hanako–to Taroo–SUB book–OBJ gave
etc.

Nonetheless, our first instincts — to attribute word order freedom to movement — might be correct, and our remorse and guilt misplaced. Kuroda (1983) reported an interesting discovery about the syntax of numeric quantification in Japanese. Japanese numbers, when they modify noun phrases, must generally occur adjacent to the phrases that they modify. Indeed, it is likely that they form a constituent with the phrase that they modify. Just as *an–* in German *anmachen* is expected to appear next to *–machen,* so one expects the *two* in *two students* to appear next to *students.* In simple sentences, this expectation is confirmed, as we can see in (5.19). The examples are carefully constructed so that we can test this expectation against easy judgments of acceptability. The **classifier** *ri* on the numeral 'two' in (5.19) agrees with *gakusei* 'student', making it clear that it is students (not pizzas) that are being counted. It is not acceptable to separate *futa–ri* from *gakusei.*

(5.19) **Japanese numerals adjacent to the phrases they modify**

a. **Gakusei–ga futa–ri** piza–o katta.
students–SUB $\text{two}_{\text{student}}$ pizza–OBJ bought
'Two students bought pizza.'

b. ***Gakusei–ga** piza–o **futa–ri** katta.
students–SUB pizza–OBJ $\text{two}_{\text{student}}$ bought

The reason for this contrast should be clear. Since the numeral and the phrase it modifies form a semantic unit, they should form a syntactic unit. When they are separated by other material as in (5.19b), they cannot form a syntactic unit, since they cannot form a constituent.

Now consider examples in which NPs are scrambled; i.e. occur in positions that do not conform to our expectations about syntactic and semantic units. Kuroda noticed something quite surprising about such cases. In sentence containing an NP that isn't where it "should" be, the numeral that modifies it has a choice as to where it is pronounced: next to where the NP *is* or next to where that NP *should be.* Thus, for instance, if one wishes to express 'A student bought two pizzas' in Japanese with *pizza* scrambled to the *left* of the subject, the expression for *two* can be found either adjacent to *pizza* or adjacent to the verb — next to where the direct object *is* or next to where it *should be.* How can we interpret these facts?

Think back on the German verb *anmachen* 'turn on'. In clauses missing a pronounced complementizer, the two pieces of this verb (a semantic unit) are nonetheless pronounced in two different places — *machen* in the complementizer position and *an* in the place where the German verb "should be" — the end of the verb phrase. This was taken as evidence for the association of the verb with two different positions, i.e. movement from V to C.

(5.20) **Evidence for scrambling as movement**

a. **Piza–o** **futa–tu** gakusei–ga katta.
pizza–OBJ two$_{\text{pizza}}$ student–NOM bought
'The student bought two pizzas.'
[numeral next to where *piza* is]

b. **Piza–o** gakusei–ga **futa–tu** katta.
pizza–OBJ student–SUB two$_{\text{pizza}}$ bought
[numeral next to where *piza* "should be"]

Kuroda's discovery has much the same character. In Kuroda's cases, the two pieces of the semantic unit *numeral* + *modified phrase* are also pronounceable in two different places. One of these places (next to the scrambled element) is unsurprising. The other (next to where the scrambled element "should be") looks like yet another instance in which a single phrase (*piza–o*) is associated with two positions, with pronunciation "spread out" over the positions. Putting it in terms of movement, we say that Japanese NPs optionally move from their expected positions (e.g. objects next to the verb) to a "scrambling position" at the left of the sentence. The syntax of Japanese numerals teaches us that scrambling is not a sign that Japanese lacks the constituent structures of English, but is, instead, an instance of movement.

Consider now the word order seen in English questions that contain a *wh*–word or phrase like *what, who, which book* or *how*. Instead of appearing in their expected position (direct object, subject, adverbial, etc.), the *wh*–element appears in a sentence–initial position which syntacticians have analysed as the same SPEC(CP) position that we noticed first in German (cf. (5.16)). In the examples below, the expected position of the italicized *wh*–phrase is represented by an underscore:[11]

(5.21) ***Wh*–phrases**

a. *What* did he get ___ for Christmas?
b. *Who* did you see ___ when you were in Tucson?
c. W*hich book* did Mary put ___ on top of the dresser?
d. H*ow* did the employees treat the workers ____?

An obvious hypothesis at this point would attribute the "unexpected" positioning of the *wh*–phrases in (5.21) to an instance of movement involving both the underscored position and the clause–initial SPEC(CP). This is, in fact, a standard hypothesis about such constructions. The movement process is called ***wh*–movement**.

[11]Evidence that that the phrase occupies SPEC(CP) comes from the similarity to German verb–second order in matrix questions. Once again, the finite verb appears in C to the left of the clause, with *what* or *who* appearing in a unique position to the left of the finite verb. As in German verb–second constructions, the auxiliary verb in main–clause questions in English is probably occupying the C position. I will not focus on this fact here, however.

Moreover, McCloskey (1995) provides evidence for *wh*–movement that is quite parallel to the evidence we have examined for V–to–C movement in German and for scrambling as movement in Japanese. McCloskey's evidence comes from a dialect of Irish English spoken in West Ulster.

West Ulster English, like many American dialects, can modify question words like *what* with *all* (*what all*). This usage makes *what* a plural.

(5.22) a. **What all** did he get ___ for Christmas?
b. **Who all** did you see ___ when you were in Derry?

McCloskey noticed that the *all* of *what all*, like the Japanese numerals, can also be pronounced in the position of the underscores — i.e. in the "expected" positions of the moved phrases. In other words, the *wh*–phrases at the front of the sentences in (5.22) also have a presence in direct object position, following the verb. This is the classic pattern of movement.

(5.23) **Evidence for *wh*–movement**
a. **What** did he get __ **all** for Christmas?
b. **Who** did you see __ **all** when you were in Derry?

It is of course necessary to establish, as we did for Japanese in (5.19b), that this pattern is really due to movement, and that speakers of West Ulster English do not just say the word *all* wherever they feel like it. This is the point of (5.24) below. These are cases in which *all* is separated from its *wh*–word but is not pronounced in the direct object positions with which the *wh*–phrases are associated:[12]

(5.24) a. ***What** did he **all** say (that) he wanted ____ to do?
b. ***Who** did he meet ____ in Derry yesterday **all**?

The impossibility of (5.24a–b) reinforces McCloskey's interpretation of (5.23a–b), and, with it, the concept of movement developed in this chapter.

Summary

We have now taken a quick look at two central phenomena of sentence grammar. We saw first how the words of sentences are hierarchically organized into phrases. We then saw that certain words and phrases occupy more than one position in syntactic structure — a phenomenon called *movement.* We were fortunate to be able to detect movement through the existence of constructions in which different parts of a moved phrase are pronounced in different positions that this phrase occupies. In the next section, we

[12]West Ulster Irish does in fact allow *all* to be pronounced to the left of complementizers in sentences in which subordinate clauses themselves contain subordinate clauses, as in ***What** did he say **all** (that) he wanted to do?*. What this means is that a phrase associated with the highest SPEC(CP) position may be associated with other SPEC(CP) positions in a syntactic tree. (This is sometimes called **successive cyclic movement**.) West Ulster English evidence for this proposal was the main point of McCloskey's paper.

examine in more detail some of the laws that govern syntactic structure and syntactic movement — asking the central question of this volume: how do these laws interact?

The story so far:

1. The units of syntax are hierarchically organized. That is, sentences have internal **structure**.
2. Some words and phrases occupy more than one position in these structures, a phenomenon called **movement**.
3. There are laws governing the fundamental organization of phrase structure, as well as movement. We have not yet discussed how these laws interact.

2 The Problem of Ineffability

In particular, we want to ask whether the pattern of constraint interaction in phonology investigated by Prince and Smolensky also governs the laws of syntax. Our survey of syntax allows us to approach this question in a fairly informed and well–grounded fashion. We can ask, for example, (1) whether the principles that distinguish proper from improper constituent structure interact in an OT manner, and (2) whether the principles that guide movement interact in an OT manner.

Ineffability Suggests a Clash & Crash *Model*

The predominant research traditions give some negative answers to these questions. In fact, there are good reasons for these negative answers. In many domains, it looks as though the laws underlying constituency and movement might not interact in an obviously OT fashion.

For example, syntacticians have studied the conditions which require *wh*–movement in English questions. They have also studied conditions which sometimes make *wh*–movement impossible. The adverb *how*, for example, can undergo *wh*–movement — the process we first looked at in West Ulster English. In (5.25) *how* has moved from a subordinate clause.

(5.25) I'm wondering *how* you think [employees should treat their subordinates __]

But movement of *how* from other types of subordinate clauses is completely impossible. For example, movement of *how* from a relative clause is not allowed. One might think of uttering a sentence like (5.26) — for example, if you want to be reminded of the regulations governing on–the–job behavior — yet the construction, with *how* understood as modifying *treat their subordinates*, is completely impossible.

Crucially, there is *no* acceptable alternative to (5.25) that involves the same words and the same interpretation. For example, a structure in which *wh*–movement simply fails to occur is also impossible.

(5.26) *I'm wondering *how* the company will fire [any employees who treat their subordinates __].

(5.27) *I'm wondering the company will fire [any employees who treat their subordinates *how*].

The standard interpretation of these facts attributes them to two constraints interacting in an apparently *non*–OT fashion:[13]

(5.28) **Obligatory *wh*–movement:** An embedded question must have a *wh*–phrase in SPEC(CP).

(5.29) **Constraint on movement:** *Wh*–movement cannot extract an adverbial from a relative clause.

When the desires of the two constraints clash, the conflict does not seem to be resolved by ranking one constraint over the other and picking an output that satisfies the more highly ranked constraint. Rather, the conflict seems to block the existence of any acceptable output — a situation called **crash** by Chomsky (1995b). We can therefore call this traditional perspective on constraint interaction **Clash & Crash**. Evidence for Clash & Crash comes from the phenomenon of **ineffability** — inputs that seem to yield no acceptable output. OT approaches, by contrast, always pick an output for any given input, since every clash is amicably resolved through constraint ranking. Now the fact of the matter is that there is indeed no way to ask the question in (5.26) and (5.27) using the means (words and structures) employed in these examples.

Ineffability appears to be a widespread phenomenon. Let us look at another example. Although we see *wh*–movement of a *wh*–phrase to SPEC(CP) accompanied by verb movement to C itself, we do not typically see *wh*–movement to SPEC(CP) accompanied by an actual complementizer pronounced in C.

(5.30) *I wonder *who* that Mary invited __ to the party.

For the moment, let us call this the ***Wh*–C Constraint**, although we will actually end up explaining (5.30) in a very different manner.[14]

Sometimes, however, the *presence* of a pronounced complementizer is essential to the well–formedness of a structure. For example, an infinitival clause only allows a

[13]These are not their standard names. The constraint requiring *wh*–movement is generally assumed to follow from a general theory about obligatory vs. impossible movement. The constraint blocking adverbial extraction is a special case of a more general constraint called the **Empty Category Principle** (ECP).

[14]The *Wh*–C Constraint is generally known as the **Doubly Filled Comp Filter**, a name that reflects some earlier analyses of the phenomenon (Keyser 1975; Chomsky & Lasnik 1977).

pronounced subject when the subject is immediately preceded by an active verb or by the prepositional complementizer *for*. In (5.33b) below, *Bill* is immediately preceded by the noun *preference*, which is the source of its unacceptability.

(5.31) ***Wh–C* Constraint:** No complementizer is pronounced in C when a phrase is pronounced in SPEC(CP).

(5.32) **a.** Sue would prefer for Bill not to tell Pete about the decision.
b. Sue would prefer Bill not to tell Pete about the decision.

(5.33) **a.** Sue's strong preference for Bill not to tell Pete about the decision.
b. *Sue's strong preference Bill not to tell Pete about the decision.

This fact probably follows from a more general requirement. This requirement, discussed further in Chapter 6, requires pronounced NPs to occupy the sort of position in which a language like Latin might make morphological case marking available. Syntacticians call this requirement the **Case Filter**. It amounts to the claim that languages without much morphology on their nouns nonetheless act as though the morphology were present.

(5.34) **Case Filter**: A pronounced NP must occupy a Case position.

The subject of an infinitive is only a case position when it is structurally close to a case–assigning element such as the prepositional complementizer *for*. This is why the pronounced subject of the infinitive in (5.33b) is unacceptable without the complementizer *for*. On the other hand, if the pronounced subject is replaced with an *un*pronounced reflexive pronoun, the result does not violate the Case Filter. The unpronounced reflexive is written *Pro*, and is the focus of extensive discussion in Chapter 6. In (5.35), its antecedent is *Sue.*

(5.35) Sue's strong preference *Pro* not to tell Pete about the decision bothers me.

Now what happens when the Obligatory W*h*–Movement Constraint, the *Wh*–C Constraint and the Case Filter impose contradictory requirements on a structure? For example, suppose we consider an embedded (subordinate) *wh*–question which is infinitival and which has a pronounced subject (not *Pro*).

- If you fail to move the *wh*–phrase, the Obligatory W*h*–Movement Constraint is violated, although *for* can then occupy C, satisfying the *Wh*–C Constraint and the Case Filter.
- If you do move the *wh*–phrase, then satisfying the W*h*–C Constraint will require not pronouncing *for*. This will violate the Case Filter.
- If you pronounce *for*, you will satisfy the Case Filter, but violate the W*h*–C Constraint.

Is this conflict resolved amiably through constraint ranking, in an OT fashion? Apparently not. No constraint seems to give way to any other. The clash among the constraints seems to lead to a situation in which no variant of the sentence can be said.

(5.36) Inviolable constraints?
a. *Mary wonders [Bill to buy which book at the store].
[violates Obligatory *Wh*–Movement]
b. *Mary wonders [which book for Bill to buy at the store].
[violates *Wh*–C Constraint]
c. *Mary wonders [which book Bill to buy at the store].
[violates Case Filter]

If the subject were the empty pronoun *Pro* (see Chapter 6) instead of *Bill*, there would be no problem, since *Pro* is not subject to the Case Filter.

(5.37) Mary wonders [which book *Pro* to buy __ at the store].

Example (5.37) is important, since some languages (e.g. German) seem to lack infinitival questions entirely. English is not such a language, which tells us that (5.36) really does teach us something special.

Ineffability and OT

Now it would be foolish to claim that one cannot account for ineffability using the mechanics of OT. In OT, the output does not have to use the same means (words and structures) as the input, but merely has to be *maximally faithful* to the input. Deviations from FAITHFULNESS occur when some constraints outrank it. An OT account of (5.36) might, for example, rank counterparts to our two constraints above FAITHFULNESS. This would favor an output that maximally satisfies both constraints and does *not* employ the same words or structures as those found in the input. The nature of this output would depend on the details of the analysis. The output might not even receive the same semantic interpretation as that which might be associated with the input. For example, the output relevant to (5.36) might be a structure in which *Bill* is replaced with *Pro*, or *to* by *should.*

Nonetheless, an effort of this sort to handle ineffability within OT would not in and of itself constitute an argument in favor of OT constraint interaction unless one could find grounds for actually *favoring* the OT approach. Do we follow the traditional dictum "If you can't say something nice, don't say anything at all" (Clash & Crash), or the less absolutist dictum "If you can't say something nice, say the best thing you can" (OT)? One wants to know the truth.

The argument can be made stronger than this, but a serious presentation requires more background than this chapter can supply and consideration of more alternatives than I have space to present. The form of the argument is quite simple. Suppose we assume a unified FAITHFULNESS constraint. Now suppose that a particular case of ineffability can be given an OT account if some group of constraints *G* is ranked higher than FAITHFULNESS. This ranking, however, entails that it should always count as more important to maximally satisfy the constraints in G than for the input to match the output. If in other cases the opposite is true — i.e. it looks more important for the input to match the output than to satisfy some constraint in G — then we have discovered an argument against an OT–internal analysis of ineffability built around interactions with the unified FAITHFULNESS constraint. Now this type of argument can always be countered by a more complex OT analysis in which (as in much work on phonology) there is not a single unified FAITHFULNESS constraint, but a family of constraints of the form FAITH(X), FAITH(Y), etc. But it is at this point that one steps back to compare the complexity of the emerging OT analysis with its Clash & Crash alternative. If the complexity of the OT analysis arises precisely from the attempt to explain a Clash & Crash phenomenon with OT tools, this fact should become apparent at this point. One of the strongest arguments of this type comes from the behavior of Case in infinitival relative clauses in English. For that discussion, see Pesetsky (to appear).

In addition, Prince & Smolensky (1993) allow for situations in which the **null parse** — i.e. an unparsed candidate — is the winner of the competition among candidates. In these circumstances, presumably, some external property of language makes the unparsed candidate unusable. This, in essence, is also a Clash & Crash explanation for ineffability, since the consequence of the OT system picking the null parse while some external system rejects the null parse is ineffability.

The current literature contains interesting attempts to work out a variety of OT approaches to problems in the theory of constituency and movement, but the actual arguments favoring these accounts over alternatives in the Clash & Crash mode remain subtle and inconclusive. There is not yet any substantial body of work that both explores the interpretation of ineffability from an OT perspective and attempts to justify that perspective in light of its obvious competitor — the claim that ineffability arises when constraints clash. (There is an implicit comparison in Legendre et al. (to appear) and in Grimshaw (to appear)). Consequently, the investigation of possible OT interactions in the areas of constituency and movement is not yet a suitable topic for a general introduction such as this.[15]

A useful tactic when considering novel ideas, is to look — not for problems that have already received satisfying solutions — but for problems that are largely unsolved. This is not as easy as it sounds. One learns to live with one's unsolved problems, and with time, one becomes so used to the unsolved that it almost comes to look solved. The existence of movement imposes on the grammar a very special burden: accounting for how and where phrases are *pronounced* when they occupy more than one position. The burden actually extends to a number of cases in which unmoved phrases show interesting patterns of pronunciation and non–pronunciation. Attempts to deal with this question in the traditional program of syntactic theory have been fairly unsuccessful, and syntacticians have become used to the unsuccessful proposals that are popular.

Interestingly, it looks very likely that the systems of grammar that deal with *this* set of questions have a strongly OT character. Could this be why traditional Clash & Crash perspectives on *this* issue, unlike the others discussed above, have explained strikingly little? Let us see.

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[15]For current collections of papers on this topic, see Barbosa et al. (to appear) and the syntax papers in Beckman et al. (1995).

Clash & Crash vs. OT:
Some laws of syntactic structure and movement seem to interact in a **Clash & Crash** fashion, rather than in an **Optimality Theoretic** manner.

3 Pronunciation of Structures with Movement

In German, Japanese and West Ulster English, we saw that the pronunciation of a moved element could be spread over more than one position. Intriguing though those examples are, they are perhaps the exception rather than the rule. For example, many (perhaps most) dialects of English that have the expression *what all* do not allow the separation of *all* from *what* in the manner of West Ulster English. Nonetheless, even when the patterns of West Ulster are unavailable, we find plentiful evidence for the full association of words and phrases with more than one position.

For example, think back on the relation between coreference and structure in examples (5.5a–b), which I reproduce as (5.38) below.

(5.38) Patterns of coreference and non–coreference

a. He criticized the picture of John. [*he* cannot be *John*]

b. His former teacher criticized the picture of John. [*his* may be *John*]

The key factor was a prohibition on pronouns that **c–command** their antecedents. Now consider the patterns of coreference found in sentences identical to (5.38) except that the direct object has undergone *wh*–movement. Surprisingly, the judgments of coreference remain unchanged.

(5.39) Patterns of coreference and non–coreference are unchanged under *wh*–movement

a. Which picture of John did he criticize?
[*he* cannot be *John*]

b. Which picture of John did his former teacher criticize?
[*his* may be *John*]

The facts are surprising because in neither case does the pronoun c–command the position in which its antecedent is pronounced. If "what you hear is what you get" holds for syntax, (5.39a) would be a puzzle. But if the *wh*–phrases at the front of the sentences in (5.5) are also associated with the direct object position, then the puzzle disappears. The occurrence of *which picture of John* in SPEC(CP) is not c–commanded by *he* in (5.39a), to be sure — but the occurrence of this phrase in direct object position *is* c–commanded by *he* in (5.39a) (and not c–commanded by *his* in (5.39b)).

(5.40) a. Which picture of John did he criticize ~~which picture of John~~?
[*he* cannot be *John*]
b. Which picture of John did his former teacher criticize ~~which picture of John~~?
[*his* may be *John*]

Given that coreference patterns in (5.39) reveal the presence of *which picture of John* in direct object position as well as in SPEC(CP), we have to ask what principles dictate that only the higher (leftmost) occurrence of this phrase gets pronounced. In the present case, the relevant principles may seem trivial. For example, if the association of the *wh*–phrase with SPEC(CP) arises via movement from direct object position (taking seriously for the moment the "sequential" property of movement), then one might conclude that pronunciation uniquely targets the *last* position occupied by a moved phrase. Alternatively, we might note that the rightmost occurrence of the *wh*–phrase in (5.40) is structurally lower than the leftmost occurrence. An occurrence of a moved phrase that is not the highest occurrence is called a **trace.** Higher occurrences of a moved phrase are called **antecedents of the trace**. We then posit a constraint that requires a trace to be unpronounced, in effect leaving a **gap** where the trace occurs.

(5.41) **SILENT TRACE:** Don't pronounce the traces of a moved constituent.

The pronunciation problem does not stop here, however. As we explore this problem further, its OT flavor begins to emerge. This is nowhere clearer than in relative clause constructions, which share many properties with *wh*–questions.

Relative Clauses in English

The problem of coreference in (5.40) can be reproduced in relative clauses as well. Patterns of coreference tell us that English relative clauses, unsurprisingly, also involve *wh*–movement.

(5.42) **Patterns of coreference and non–coreference in relative clauses**
a. Mary, whose picture of John he criticized __, is now a well–known photographer.
[*he* cannot be *John*]
b. Mary, whose picture of John his former teacher criticized __, is now a well–known photographer.
[*his* may be *John*]

The data in (5.42) are straightforwardly explained if *wh*–phrases are associated with the direct object position, just as they were in (5.40). SILENT TRACE dictates that the italicized material is not pronounced.

(5.43) a. Mary, whose picture of John he criticized ~~whose picture of John~~,...
b. Mary, whose picture of John his former teacher criticized ~~whose picture of John~~,...

In (5.42), the moved phrase is complex. The *wh*–word *whose*, which is linked to *Mary* (its antecedent), is a subpart of the phrase that undergoes *wh*–movement. Suppose we now examine simpler relative clauses in which the *wh*–word linked to the head *is* the phrase that undergoes movement. We suddenly see a greater variety of pronunciation patterns than we have seen so far. In particular, though the trace remains unpronounced, the *wh*–word need not be. Furthermore, when the *wh*–word is not pronounced, the complementizer *that* may be. In fact, the only *im*possible configuration is the one that violates the *Wh*–C Constraint (since both the *wh*–element in SPEC(CP) and the complementizer *that* are pronounced).

(5.44) Simple relative clauses in English: three options are available
- **a.** *the person [*whom that* Mary invited __ to the party]
- **b.** the person [*whom* ~~*that*~~ Mary invited __ to the party]
- **c.** the person [~~*whom*~~ *that* Mary invited __ to the party]
- **d.** the person [~~*whom*~~ ~~*that*~~ Mary invited __ to the party]

Relative clauses in which the *wh*–word is embedded in a more complex structure do not have this multiplicity of options. For example, in (5.42) the *wh*–phrase in SPEC(CP) must be pronounced. Nor can a PP in SPEC(CP) be unpronounced:[16]

(5.45) Complex relative clauses in English: one option
- **a.** *the person [*to whom that* Mary spoke __ at the party]
- **b.** the person [*to whom* ~~*that*~~ Mary spoke __ at the party]
- **c.** *the person [~~*to whom*~~ *that* Mary spoke __ at the party]
- **d.** *the person [~~*to whom that*~~ Mary spoke __ at the party]

An obvious difference between the *wh*–phrase in (5.44) and the *wh*–phrase in (5.45) is the fact that the head of the relative furnishes an antecedent for the entirety of the *wh*–phrase in (5.45), but not in (5.25). In (5.25), *person* is an antecedent for *whom*, but not for the larger phrase *to whom*.

In general, antecedentless phrases with semantic content like *to whom* in (5.45) cannot be unpronounced. This fact is accounted for by a principle called the Recoverability Condition — the idea being that the semantic content of elements that are not pronounced must be recoverable from local context. I name the corresponding OT contraint **RECOVERABILITY**; this constraint does not force pronunciation of semantically contentless words like the complementizer *that*, as in (5.46b), or of pronouns with antecedents like *whom*.

[16]The preposition can also be **stranded**, yielding semantically equivalent relative clauses like *the person who Mary spoke to __ at the party*. I will assume that the question of whether the preposition does or does not move to SPEC(CP) is a matter of movement, not of pronunciation. In more complex cases (infinitival relatives), one can actually argue for this claim, since any other assumption would fail to predict the optionality of stranding vs. not stranding the preposition.

(5.46) **The complementizer in English declarative sentences**
  a. Mary thinks *that* Peter is hungry.
  b. Mary thinks ~~*that*~~ Peter is hungry.

The Optimality–Theoretic Character of French Relative Clauses

We have not yet seen OT in action, but we will, once we compare English relative clauses to relative clauses in French.[17] The syntax of French embedded clauses looks a lot like English. It is often particularly illuminating to study languages like English and French that differ fairly minimally. When two languages differ minimally, the differences that do exist stand out in sharper relief, and we are less likely to posit nonexistent connections among phenomena.

Interestingly, in tensed embedded clauses, the most salient differences between French and English lie in the patterns of allowable pronunciations and non–pronunciations. In particular, French is uniformly more restrictive. For example, the complementizer *que* in an embedded declarative must be pronounced, in contrast to English. That is, the alternation in (5.46) is not found in French.

(5.47) **The complementizer in French declarative sentences**
  a. Je crois [$_{CP}$ que Pierre a faim].
   I believe that Pierre is hungry.
  b. *Je crois [$_{CP}$ ~~que~~ Pierre a faim].

From a Clash & Crash perspective, one might say that French simply has a constraint that requires C to be pronounced. We will see shortly, however, that this would not be an accurate description of French.

Another obvious difference between English and French is found in relative clauses. In relative clauses with a *wh*–phrase that can be recoverably unpronounced, English allows the three patterns of pronunciation in (5.44): a ***wh*–relative** (with C unpronounced), a ***that*–relative** (with the *wh*–word unpronounced), and a ***zero*–relative** (with both unpronounced). In contrast, French relative clauses have only one pronunciation option: the *that* relative. All other possibilities are strongly unacceptable:[18]

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[17]Much of what I will say about French is true of other Romance languages, such as Italian (Cinque 1981).

[18]The French data are confounded by the existence of an entirely different word *qui* (which I write *qui*$_2$), which is the shape taken by the complementizer *que* when the closest subject has undergone *wh*–movement:

(i) la table qui$_2$ est dans ma chambre
  the table which is in my room

Qui$_2$ has properties quite unlike the *qui* discussed in the text. For example, it is compatible with the relativization of inanimate NPs, unlike the *qui* discussed in the text, which is strictly animate (and translates English *who*). Also, it appears as an alternate of *que* introducing simple declarative clauses from which the subject has undergone *wh*–movement:

(ii) Quelle table penses–tu [qui$_2$ est dans ma chambre]?
  Which table think you that is in my room
  'Which table do you think is in my room?'

(5.48) **Simple relative clauses in French: one option**

a.	*l'homme [$_{CP}$ qui que je connais]	(*the man who that I know)
b.	*l'homme [$_{CP}$ qui ~~que~~ je connais]	(the man who I know)
c.	l'homme [$_{CP}$ ~~qui~~ que je connais]	(the man that I know)
d.	*l'homme [$_{CP}$ ~~qui~~ ~~que~~ je connais]	(the man I know)

What lies behind these two differences between English and French? I would like to suggest that these differences have a common origin. The problems of (5.47) and (5.48) display an obvious similarity. In each case, the only acceptable pattern of pronunciation is one which makes *que* the first pronounced word of the embedded clause (CP). Suppose, then that the patterns in (5.47) and (5.48) are due to a constraint favoring just this situation, which I call **LEFTEDGE(CP).**[19]

(5.49) **LEFTEDGE(CP):** The first (leftmost) pronounced word in CP must be the complementizer.[20]

It is LEFTEDGE(CP) that favors pronunciation over non–pronunciation of *que*, and it is LEFTEDGE(CP) that favors non–pronunciation over pronunciation of relative *wh*.

Now let us ask what happens when LEFTEDGE(CP) clashes with RECOVERABILITY. Is the class resolved, in the manner of Optimality Theory, by allowing an output that violates one of these two constraints, or does the class produce a crash, i.e. no acceptable output? In fact, this interaction, unlike those we have examined so far, is resolved in a strikingly optimality–theoretic fashion. When the *wh*–phrase in SPEC(CP) of a relative clause cannot be unpronounced in conformity with recoverability, it is impossible to satisfy LEFTEDGE(CP). The system does not yield a Crash, but tolerates the violation of LEFTEDGE(CP) in favor of satisfying RECOVERABILITY. In particular, the *wh*–phrase whose pronunciation prevents CP from beginning with *que* is pronounced, with no loss

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Additionally, qui_2 can cooccur with a literary use of *qui* in the constructions discussed in the Section 5. Throughout this paper, I ignore relativization of subjects, to avoid discussing qui_2. Subject relativization in English also has special properties, which I ignore here.

[19]In Pesetsky (to appear), I argue that LEFTEDGE(CP) more generally requires the first pronounced word in a clause to be one of the "functional" words on the path from V to C — not necessarily C itself. LEFTEDGE(CP) as discussed in this chapter looks like a generalized alignment constraint (see Russell this volume). It turns out, however, that LEFTEDGE(CP) lacks the "gradient" property of alignment constraints explored in the Generalized Alignment literature. All examples that do not totally satisfy LEFTEDGE(CP) seem to be equally starred; it is no better to find the relevant sort of function word in second position within CP than to find it in tenth position. (The evidence for this point falls outside the scope of this chapter.) The difference between LEFTEDGE(CP) and alignment constraints may reflect an inadequacy of LEFTEDGE(CP) as formulated here, or it may be telling us something interesting.

[20]In main–clause declarative sentences, the complementizer is not found. That is, in French, simple sentences do not begin with *que* and in English simple sentences are not introduced by *that*. This fact may be a indication of some other constraint that overrules LEFTEDGE(CP) in main clauses, or it may be a property of the system that provides constituent structures.

of acceptability. That is, LEFTEDGE(CP) is ranked below RECOVERABILITY and is violable in classic OT fashion.

(5.50) RECOVERABILITY » LEFTEDGE(CP)

On the other hand, something interesting does happen in these relative clauses that is not explained by RECOVERABILITY and LEFTEDGE(CP) alone: the complementizer itself is unpronounced, conforming to the observation that we earlier called the *Wh–C* Constraint. That is, only (5.51b) is possible.

(5.51) Complex relative clauses in French: one option

a. *l'homme [$_{CP}$ avec qui que j'ai dansé]
*the man with whom that I danced

b. l'homme [$_{CP}$ avec qui ~~que~~ j'ai dansé]
the man with whom I danced

c. *l'homme [$_{CP}$ ~~avec qui~~ que j'ai dansé]
*the man that I danced

d. *l'homme [$_{CP}$ ~~avec qui~~ ~~que~~ j'ai dansé]
*the man I danced

RECOVERABILITY and LeftEdge(CP) by themselves would allow both (5.51a) and (5.51b) to survive as winning candidates. The tableau below and elsewhere compares possible pronunciations of a given structure. (See the final section of this chapter for discussion.)

(5.52) Complex relative clauses in French: an incomplete result

	Candidates	RECOV	LE(CP)
☞	*l'homme [$_{CP}$ avec qui que j'ai dansé] the man with whom that I danced		*
☞	l'homme [$_{CP}$ avec qui ~~que~~ j'ai dansé		*
	*l'homme [$_{CP}$ ~~avec qui~~ que j'ai dansé]	*!	
	*l'homme [$_{CP}$ ~~avec qui~~ ~~que~~ j'ai dansé]	*!	*

The impossibility of the first candidates in (5.51) and (5.52) is, of course, already familiar to us as the *Wh–C* Constraint. We will now try to explain this constraint. Clearly, the obligatory non–pronunciation of *que* in (5.51) is a fate that befalls *que* when other factors make it impossible for *que* to be pronounced first in its clause. In other words, there exists a constraint which (unlike LEFTEDGE(CP)) favors *non–pronunciation* of *que*. This constraint is ranked lower than LEFTEDGE(CP). Such a constraint can have an effect on output only when RECOVERABILITY prevents LEFTEDGE(CP) from being satisfied. I suggest that the constraint that forces *que* to be unpronounced in (5.51) more generally favors non–pronunciation of all closed–class function words — including complementizers, English *to* and others. Consequently, I call the constraint **TELEGRAPH**.

(5.53) **TELEGRAPH:** Do not pronounce function words (e.g. complementizers).

Ranked high, such a constraint would produce "telegraphic" speech of the sort actually found among children in their second year (examples from Radford 1990).

(5.54) Go nursery...Lucy go nursery. (Stevie, 25 months)

(5.55) Where girl go? (Claire, 24 months)

About examples of this sort, Brown & Fraser (1963) commented "[T]he striking fact about the utterances of the younger children, when they are approached from the vantage point of adult grammar, is that they are almost all classifiable as grammatical sentences from which certain morphemes have been omitted." It is tempting to see this a sign that TELEGRAPH is present in the child, ranked very high.

In adult French, however, TELEGRAPH must be ranked below LEFTEDGE(CP).

(5.56) RECOVERABILITY » LEFTEDGE(CP) » TELEGRAPH

This is not obvious from inspection of complex relative clauses alone, as (5.57) makes clear. Here, TELEGRAPH could be ranked higher than LEFTEDGE(CP), with no difference in the result.

(5.57) **Complex relative clauses in French**

	Candidates	RECOV	LE(CP)	TEL
	*l'homme [$_{CP}$ avec qui que j'ai dansé] the man with whom that I danced		*	*!
☞	l'homme [$_{CP}$ avec qui ~~que~~ j'ai dansé]		*	
	*l'homme [$_{CP}$ ~~avec qui~~ que j'ai dansé]	*!		*
	*l'homme [$_{CP}$ ~~avec qui~~ ~~que~~ j'ai dansé]	*!		

What motivates the ranking LEFTEDGE(CP) » TELEGRAPH is the fact that marking the complementizer unpronounced is something you do *only* when RECOVERABILITY prevents the CP from actually starting with a pronounced complementizer. In all other cases, the CP must start with the pronounced complementizer, as we saw in both (5.47) and (5.48).

(5.58) **Declarative *que***

	Candidates	RECOV	LE(CP)	TEL
☞	Je crois [$_{CP}$ que Pierre a faim]. I believe that Pierre is hungry]			*
	*Je crois [$_{CP}$ que Pierre a faim].		*!	

(5.59) Simple relative clauses in French

	Candidates	RECOV	LE(CP)	TEL
	*l'homme [$_{\text{CP}}$ qui que je connais] the man who that I know		*!	*
	*l'homme [$_{\text{CP}}$ qui que je connais]		*!	
☞	l'homme [$_{\text{CP}}$ qui que je connais]			*
	*l'homme [$_{\text{CP}}$ qui que je connais]		*!	

It is also worth noting that embedded questions behave just like complex relative clauses. This is because the *wh*–phrase in a question has no antecedent, and therefore cannot be recoverably unpronounced.

(5.60) Embedded questions in French

	Candidates	RECOV	LE(CP)	TEL
	*Je me demande [$_{\text{CP}}$ quand que Pierre arrivera]. I wonder when that Pierre will arrive		*	*!
☞	Je me demande [$_{\text{CP}}$ quand ~~que~~ Pierre arrivera].		*	
	*Je me demande [$_{\text{CP}}$ ~~quand~~ que Pierre arrivera].	*!		*
	*Je me demande [$_{\text{CP}}$ ~~quand que~~ Pierre arrivera].	*!	*	

The interaction between LEFTEDGE(CP) and TELEGRAPH not only yields the effects of what we called the *Wh*–C Constraint, but also can be said to partially *explain* it. What is explained is the environment of the effect: the fact that it is when a pronounced element precedes the complementizer within CP that something noteworthy happens. This is explained insofar as the effect arises from LEFTEDGE(CP), which has been independently motivated as an account of other phenomena. In fact, the absence of a pronounced complementizer with verb–second order in German — a central fact in the previous section — can itself be explained as a consequence of LEFTEDGE(CP) and TELEGRAPH, if the verb in second position is placed to the left of C. Unexplained, of course, is why TELEGRAPH exists in the first place. Nonetheless, evidence that TELEGRAPH does exist seems to be plentiful.

French relative clauses are the focus of this section because they are a phenomenon that suggests Optimality Theory at every turn. Consider, for example the most natural Clash & Crash treatment of the obligatory pronunciation of *que* in declarative clauses. One would posit a constraint against non–pronunciation of *que* in such a framework. This assumption would immediately run afoul of the non–pronunciation of *que* in relative clauses like (49). Consider now the most natural Clash & Crash treatment of (5.48). One might posit, for example, an obligatory rule of *wh*–deletion. This rule, in turn, would run afoul of the obligatory *non*–deletion of the *wh*–phrases in (5.51). One could always formulate the deletion rule so as to distinguish between simple and complex *wh*–phrases (just as one could always formulate a treatment of ineffability using OT constraint interaction) — but this would miss the point. Non–pronunciation is obligatory so long as recoverability is satisfied. This, of course, is a characteristic argument for OT interactions among constraints. Indeed, discussion of French relative clauses in the 1970s such as Kayne (1977) often used the phrase "deletion obligatory *up to recoverability*" to describe the patterns of pronunciation found in this construction. Sadly, researchers of the time did not look more deeply into the OT interactions they had discovered, and the topic languished for two decades.

4 French and English

As I have stressed throughout this chapter, one can use the language of Clash & Crash or OT to provide a description of most phenomena. We hope for something better — to discover the truth of the matter — how *is* our language faculty organized? But comparison of approaches to problems from different perspectives is not easy. There will rarely be a "knock–down" argument. Instead, one looks for somewhat nebulous distinctions like "the most natural proposals within framework X work better than the most natural proposals within framework Y", and one looks more generally at which framework advances the fastest and most effectively across fields of previously unsolved problems. Additionally, one hopes that evidence for a particular proposal will converge on that proposal from more than one direction. This is not a logical requirement for a true theory, but when it happens, it serves to boost one's confidence that the ideas on the table vaguely resemble the truth.

A theory like OT in which the laws of language are ranked offers the possibility that variation among languages can be attributed in some cases to differences in constraint ranking. If the sorts of laws natural to OT, when ranked and reranked, yield the set of actual grammars (and if a comparable demonstration is not possible using a Clash & Crash approach), we have a piece of converging evidence in favor of the OT approach. Few such comprehensive demonstrations exist, of course, but we can at least ask whether the signs look promising.

Suppose, for example, that TELEGRAPH were ranked higher than LEFTEDGE(CP) in some language. This sort of ranking would simply yield a language in which C is always unpronounced. Languages of this sort exist. Chinese, for example, lacks a pronounced complementizer in subordinate declarative clauses (though it has morphemes in other environments that may be complementizers; see Cheng 1991).

Tied Constraints

A more complex result would arise if TELEGRAPH and LEFTEDGE(CP) could be *tied* — i.e. if the output of LEFTEDGE(CP) and TELEGRAPH were the *union* of the outputs of the two possible rankings: i.e. the union output of LEFTEDGE(CP) » TELEGRAPH and the output of TELEGRAPH » LEFTEDGE(CP). In fact, that is exactly the case in English.[21]

[21]Actually, several types of interaction among constraints might merit the name *tie*. The one proposed here is the one that seems correct in general, though the arguments favoring it over its competitors lie outside the scope of this chapter.

First, in subordinate declarative clauses, the complementizer *that* may be pronounced (satisfying LEFTEDGE(CP) but violating TELEGRAPH) or it may be unpronounced (satisfying TELEGRAPH, but violating LEFTEDGE(CP). Second, in simple relative clauses, there are the three patterns noted in (5.44). In a *that*–relative, the *wh*– is unpronounced and *that* is pronounced. This satisfies LEFTEDGE(CP) but violates TELEGRAPH. Next, in a *wh*–relative, *that* is unpronounced and *wh*– is pronounced. This satisfies TELEGRAPH but violates LEFTEDGE(CP). Finally, in a *zero*–relative, *that* is unpronounced and *wh*– is also unpronounced. This satisfies TELEGRAPH just as well as the *wh*–relative does, and violates LEFTEDGE(CP) just as badly as the *wh*–relative does.[22]

The fourth possible pattern, in which both *wh*– and *that* are pronounced, satisfies neither LEFTEDGE(CP) (since the relative clause does not begin with *that*) nor TELEGRAPH (since *that* is pronounced). It is therefore forbidden in English just as it was in French. These patterns can be represented with tableaux, but the reader has to bear in mind that the output of the tie between LEFTEDGE(CP) and TELEGRAPH is the union of the outputs of LEFTEDGE(CP) » TELEGRAPH and TELEGRAPH » LEFTEDGE(CP). In (5.61), the first candidate is the winner on the ranking LEFTEDGE(CP) » TELEGRAPH, while the second candidate is the winner on the ranking TELEGRAPH » LEFTEDGE(CP).

(5.61) Declarative complementizer *that*

	Candidates	RECOV	LE(CP)	TEL
☞	I believe [$_{CP}$ that Peter is hungry].			*
☞	I believe [$_{CP}$ ~~that~~ Peter is hungry].		*	

In (5.62), which displays relative clauses with a simple *wh*–phrase, the third candidate (the *that*–relative) is the winner on the ranking LEFTEDGE(CP) » TELEGRAPH, while the second and fourth candidates (the two candidates in which *that* is unpronounced) are the winners on the ranking TELEGRAPH » LEFTEDGE(CP). Notice that the first candidate — the candidate that violates what we earlier called the *Wh*–C Constraint — is the winner on neither ranking. Thus we can continue to maintain that the *Wh*–C Constraint is *explained* by the OT system outlined here.

(5.62) Simple relative clauses in English

	Candidates	RECOV	LE(CP)	TEL
	*the person [$_{CP}$ who that Mary invited t to the party]		*	*!
☞	the person [$_{CP}$ who ~~that~~ Mary invited t to the party]		*	
☞	the person [$_{CP}$ ~~who~~ that Mary invited t to the party]			*
☞	the person [$_{CP}$ ~~who that~~ Mary invited t to the party]		*	

Now let us turn to relative clauses with a complex *wh*–phrase. In English, as in French, RECOVERABILITY excludes non–pronunciation of the *wh*–phrase. This means that LEFTEDGE(CP) simply cannot be satisfied by any pattern of pronunciation that satisfies

[22]In a variety of contexts, including relative clauses built on a subject (*the person who/that arrived late*) and relative clauses not adjacent to their heads, the zero option is impossible. I do not discuss this additional restriction here.

RECOVERABILITY. When no candidate satisfies a constraint in OT, the constraint simply fails to have any winnowing effect on the candidate set. The union of the outputs of LEFTEDGE(CP) » TELEGRAPH and TELEGRAPH » LEFTEDGE(CP) in these cases is thus the same as the output of TELEGRAPH alone — namely, a candidate in which the *wh–* is pronounced but the complementizer *that* is not. This is the same output as in equivalent French relative clauses.

(5.63) Complex relative clauses in English

	Candidates	RECOV	LE(CP)	TEL
	*the person [$_{CP}$ to whom that Mary spoke t at the party]		*	*!
☞	the person [$_{CP}$ to whom ~~that~~ Mary spoke t at the party]		*	
	*the person [$_{CP}$ ~~to whom~~ that Mary spoke t at the party]	*!		*
	*the person [$_{CP}$ ~~to whom that~~ Mary spoke t at the party]	*!		

Thus, a small difference between English and French — the difference between strict ranking and a tie among two constraints — accounts for a cluster of empirical differences between the two languages. If we can accumulate other results of this kind, they will constitute additional evidence for OT patterns of constraint interaction in this corner of syntax.

Pronunciation of Traces

Other phenomena connected with the pronunciation of syntactic units also participate in the system I am describing. Consider, for example, the role of SILENT TRACE in (5.41), the principle that requires that the trace of a moved constituent is not pronounced (i.e. that movement leaves a gap). SILENT TRACE outranks LEFTEDGE(CP) and TELEGRAPH in French. Otherwise, in a complex relative clause like those in (5.51), LEFTEDGE(CP) would favor a candidate in which the *trace* is pronounced instead of SPEC(CP). Such a pronunciation would be favored because it allows *que* to be pronounced first, while still satisfying RECOVERABILITY.

(5.64) *l'homme [$_{CP}$ ~~avec qui~~ que j'ai dansé *avec qui*]

At this point, we should ask whether SILENT TRACE is a violable or inviolable constraint — i.e. whether all traces are gaps. In fact, not all traces are gaps. In certain circumstances, traces are pronounced, but the pronunciation is not necessarily what we might expect.

There is a class of phrases (called **islands**) which impose special restrictions on *wh–* movement and similar phenomena. In particular, an island may not contain a gap whose antecedent is not also contained by the island. For example, a coordinate structure such as *Mary and John* cannot contain a gap whose antecedent lies outside the gap.[23]

(5.65) *This is the guy *who* I thought that [[Mary and __]] were going to the movies].

[23]I use double square brackets [[...]] to delimit an island.

It is sometimes argued that movement itself cannot link a position inside an island with a position outside an island. But in fact, movement of this sort *is* possible, so long as the trace within the island is not a gap, but is actually pronounced. On the other hand (in languages like English, at least) a trace in an island is not pronounced as a full *wh*–phrase, but is instead pronounced as a *pronoun.* A pronoun of this sort (a pronunciation of a trace) is called a **resumptive pronoun**. Resumptive pronouns are especially common in conversational English.[24] They function, for example, as the sole legitimate pronunciation of trace when *wh*–movement has taken place out of a coordination, when *wh*–movement involves a subject of a clause that is itself a question, or when *wh*–movement of a noun phrase involves a relative clause within a relative clause.[25]

(5.66) **a.** *This is the guy *who* I thought that [[Mary and __]] were going to the movies. (=(5.64))
b. This is the guy *who* I thought that [[Mary and *him*]] were going to the movies.

(5.67) **a.** *This is the guy *who* I wondered [[whether __ was going to the movies]].
b. This is the guy *who* I wondered [[whether *he* was going to the movies]].

(5.68) **a.** *There is one worker *who* the company fired the employee [[that treated __ badly]].
b. There is one worker *who* the company fired the employee [[that treated *him* badly]].

Moreover, resumptive pronouns (in English, at least) are largely *limited* to configurations involving islands. Simpler instances of *wh*–movement do not allow resumptive pronouns.

(5.69) *This is the guy *who* I like *him*.

To summarize, we saw that island phenomena can be attributed to constraints that prohibit gaps in various contexts.[26] We then noted that resumptive pronouns — an alternative to gaps — are possible if and only if they are necessary as a way of avoiding island violations. But this is an absolutely classic OT interaction. Obviously, the constraint that *dis*favors resumptive pronouns (SILENT TRACE) is violable, and is ranked lower than the island constraints that disfavor gaps The lower–ranked constraint (as always in OT) is violated whenever that is the only way to satisfy the higher–ranked constraint.

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[24]Resumptive pronouns are, in fact, *limited* to conversational styles of English. I return to this important fact in Section 5.

[25]The similarity of this configuration to that in (5.26) is discussed in the Section 5.

[26]The quest for a *unified* account of island phenomena has always been a lively topic of syntactic research. In this chapter, I sidestep the question of how many distinct island conditions there might be, using the vague term **islands** to refer to the collection of constraints that explain island effects. The proper analysis of island effects is tightly bound to the question, also sidestepped here, of sequential vs. non–sequential views of movement phenomena.

Furthermore, OT immediately gives us a way of understanding why these constructions show *pronouns* rather than full *wh*–phrases. If pronouns are a way of pronouncing a *wh*–phrase which counts as "more silent", then SILENT TRACE is less violated by pronouncing the trace as a pronoun, than by pronouncing the trace in its full glory as *wh*–phrase. This makes sense if pronouns are bundles of grammatical features without any notional properties. Pronouncing a category as a pronoun means giving phonological shape to one portion of its properties, but not another. Since OT tolerates constraint violation, but minimizes such violation, SILENT TRACE — when it simply must be violated — will prefer a pronunciation that targets only one portion of the properties of trace (i.e. a pronoun) over full pronunciation of the trace.

The proposal is schematized in tableau (5.70). I assume that one star is assessed against SILENT TRACE for pronouncing *anything* in trace position, and an extra star for examples in which the trace is fully pronounced:

(5.70) Island violations saved by resumptive pronouns

	Candidates	ISLANDS	SILTRC
	*the guy who I thought that Mary and __ were going...	*!	
☞	the person who I thought that Mary and him were going ...		*
	*the person who I thought that Mary and whom were going ...		**!

The idea that pronouns are a partial pronunciation of a fuller phrase resonates with traditional ideas about pronouns, but has far–reaching consequences that I do not explore in this chapter. I simply offer the case of resumptive pronouns as another example of OT interactions in sentence pronunciation.

Here too, variation among dialects may be explainable as a consequence of constraint reranking. This view might shed light on some data concerning resumptive pronouns in the French of young children studied by Labelle (1990), to which we turn next.

Child French

Young French–speaking children display a pattern of relativization that differs from what I have described for adults.[27] In particular, they differ in cases like (5.51), repeated as (5.71a) below, in which a prepositional *wh*–phrase cannot be unpronounced in SPEC(CP) due to RECOVERABILITY. I have indicated not only the unpronounced complementizers but also the unpronounced traces. Labelle elicited relative clauses from 108 French–speaking children, 3 to 6 years old, living in the Ottawa area. The experimental stimuli were designed so as to elicit a variety of relative clauses. (Children were telling the experimenter about the places they would like to put various stickers.)

Among the 1,348 relative clauses elicited by Labelle, *not one child* produced a relative clause like those in (5.71), in which a prepositional *wh*–phrase is pronounced in SPEC(CP), necessitating the non–pronunciation of complementizer *que*. Instead, they employed various other strategies. In some cases, they left the prepositional *wh*–phrase

[27]Labelle's results may actually reflect an adult norm in the Quebec speech communities where she did her research. This would transform our discussion from an observation about language acquisition to one about dialect variation.

entirely unpronounced, in violation of RECOVERABILITY, but permitting the satisfaction of LEFTEDGE(CP), as in (5.72). In other cases, the prepositional *wh*–phrase was pronounced in its trace position as a resumptive pronoun, as in (5.73).[28]

(5.71) **SILENT TRACE and complex relative clauses: adult French**

a. l'homme $[_{CP}$ avec qui ~~que~~ j'ai dansé ~~avec qui]~~
the man with whom ~~that~~ I danced ~~with whom~~

b. l'homme $[_{CP}$ à qui ~~que~~ le papa montre un dessin ~~à qui]~~
the man to whom ~~that~~ the dad shows a picture ~~to whom~~

c. la boîte $[_{CP}$ dans laquelle ~~que~~ le camion rentre ~~dans laquelle]~~
the box in which ~~that~~ the truck goes ~~in which~~

(5.72) **RECOVERABILITY violations in child French**

la boîte $[_{CP}$ ~~dans laquelle~~ que la petite fille elle embarque ~~dans laquelle]~~
the box ~~in which~~ ~~that~~ the little girl she goes ~~in which~~
(K 4 years, 4 months)

(5.73) **SILENT TRACE and complex relative clauses: child French**

a. l'homme $[_{CP}$ à qui ~~que~~ le papa lui montre un dessin]
the man to whom ~~that~~ the dad to–him shows a picture (cf. (5.71b))
(JF, 5 years, 0 months)

b. la boîte $[_{CP}$ dans laquelle ~~que~~ le camion rentre dedans]
the box in which ~~that~~ the truck goes in it (cf. (5.71c))
(S, 4 years, 8 months)

Do these examples violate RECOVERABILITY by failing to pronounce the complex *wh*–phrase in SPEC(CP)? Probably not. RECOVERABILITY cares only that a phrase must be pronounced under certain circumstances. It does not care about *which position* that phrase is pronounced in, when it is associated with more than one position.[29] But the examples do violate SILENT TRACE.

Why are they allowed to violate SILENT TRACE? No islands are involved in these examples, after all. One very tempting answer involves differences in the ranking of SILENT TRACE with respect to LEFTEDGE(CP) in the adult grammar and in the children's grammar.

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[28]There was a third pattern, in which the resumptive is a full NP duplicating the head of the relative.

(i) la boîte que la petite fille est debout sur la boîte
the box that the little girl is standing on the box

This example is unexplained if relative clauses involve movement of a *wh*–element whose trace is partially pronounced in configurations where SILENT TRACE is violated. See Guasti and Shlonsky (1995) for some discussion.

[29]In some cases, preposition + pronoun is represented in French by a single word, e.g. *lui* 'to him' and *dedans* 'in it'.

If SILENT TRACE is ranked higher than LEFTEDGE(CP), it is more important to avoid resumptive pronouns than to satisfy LEFTEDGE(CP), and cases in which a prepositional phrase undergoes *wh*–movement indeed violate LEFTEDGE(CP). That is the adult pattern.

(5.74) Adult French pattern: SILENT TRACE » LEFTEDGE(CP)

l'homme à qui que le papa montre un dessin *à qui* the man to whom that the dad shows a picture *trace*	RECOV	SILENT TRACE	LE(CP)	TEL
[$_{CP}$ à qui que ... gap]			*	*!
☞ [$_{CP}$ à qui ~~que~~ ... gap]			*	
[$_{CP}$ ~~à qui~~ que ... gap]	*!			*
[$_{CP}$ ~~à qui que~~ ... gap]	*!			*
[$_{CP}$ à qui que ... P pronoun]		*!	*	*!
[$_{CP}$ à qui ~~que~~ ... P pronoun]		*!	*	
[$_{CP}$ ~~à qui~~ que ... P pronoun]		*!		*
[$_{CP}$ ~~à qui que~~ ... P pronoun]		*!	*	*

What about the children's pattern? The children's dialect seems to ensure that relative clauses start with a pronounced complementizer in *all* cases. This is just what we expect if the children's grammar, unlike the adult's, ranks SILENT TRACE *lower* than LEFTEDGE(CP). This ranking makes it more important to satisfy LEFTEDGE(CP) than to satisfy SILENT TRACE. A resumptive pronoun is the minimal violation of SILENT TRACE necessary to allow non–pronunciation of the *wh*–phrase in SPEC(CP). That is the children's pattern (5.75).[30]

Finally, if individual children allow both (5.72) and (5.73), they may be displaying a tie between RECOVERABILITY and SILENT TRACE. Example (5.72) satisfies SILENT TRACE but violates RECOVERABILITY. Example (5.73) satisfies RECOVERABILITY (if our discussion just now was correct), but violates SILENT TRACE. A tie between these constraints will allow both patterns.

These facts about child French are another demonstration of how minor differences in ranking seem to correlate with observed variation in linguistic knowledge and use. Significantly, the result depends on the idea that the constraints which govern pronunciation of structures with movement interact in an optimality–theoretic manner. The result is far from conclusive, of course — but it is a start. Among the many questions we need to understand is why the children's grammar systematically differs from the adult grammar in the way it does. Is there a default ranking of constraints? If so, is this a biologically rooted property of the system, or a response to some complex property of the

[30]However, the children in Labelle's study did not avoid constructions with a prepositional phrase in SPEC(CP). For example, questions elicited from the children did have this form.

(i) Sur quoi on pèse? [JF, 2 years, 0 months]
on what one pushes
'What does one push on?'

I do not attempt to explain the difference between questions and relative clauses here. It may relate to the observation that declarative main clauses are not introduced by *que*. It makes sense that LEFTEDGE(CP) would not disfavor pronunciation of prepositional phrases like *sur quoi* in SPEC(CP) if there is some independent reason for LEFTEDGE(CP) not to prevail in main clauses.

data to which the children are exposed? Answers to these questions and others have yet to be given.

(5.75) Child French pattern: LEFTEDGE(CP) » SILENT TRACE

	l'homme à qui que le papa montre un dessin *à qui* the man to whom that the dad shows a picture *trace*	RECOV	LE(CP)	SILENT TRACE	TEL
	[$_{CP}$ à qui que ... gap]		*!		*!
	[$_{CP}$ à qui ~~que~~ ... gap]		*!		
	[$_{CP}$ ~~à qui~~ que ... gap]	*!			*
	[$_{CP}$ ~~à qui~~ ~~que~~ ... gap]	*!			*
	[$_{CP}$ à qui que ... P pronoun]		*!	*	*!
	[$_{CP}$ à qui ~~que~~ ... P pronoun]		*!	*	
☞	[$_{CP}$ ~~à qui~~ que ... P pronoun]			*	*
	[$_{CP}$ ~~à qui~~ ~~que~~ ... P pronoun]		*!	*	*

5 On the Nature of Things

What bigger picture emerges from this discussion? We began with two fundamental facts about syntax: the organization of words and phrases into constituent structures, and the occasional association of these words and phrases with more than one position in structure. Clearly the grammar has systems of principles that determine constituent structure and the patterns of multi–association (movement) within it. We then observed that these two systems implicate a third system: a group of principles that determines how words and phrases are pronounced, in particular when they are associated with more than one structural position.

When the facts of system were laid out in this manner, we easily saw that the principles of this third system have an OT character. At the very beginning of this chapter, I asked whether this sort of conclusion could extend to the other two systems. We already saw some reasons to give a negative answer to this question. In the terms of Prince & Smolensky (1993), this negative answer would mean that structure and movement fall under GEN, not under EVAL. In essence, it looks as though the system of constraints governing pronunciation, though organized internally along OT lines, interacts with other parts of grammar in a Clash & Crash fashion. The winning candidate of the pronunciation system, for example, when it violates a constraint outside this system, can create a situation in which *nothing* is sayable — a situation of ineffability.

Recall, for example, what happened when the requirements of the *Wh*–C Constraint clashed with the Case Filter. The result was ineffability — a crash. As the discussion progressed, I suggested that the *Wh*–C Constraint is not an independent principle of language, but a consequence of LEFTEDGE(CP) and TELEGRAPH within the pronunciation system of syntax. Putting these two conclusions together, we can see that the pronunciation system — though its *internal* organization is optimality theoretic — must interact with the Case Filter in a Clash & Crash fashion. That is, internal affairs of some pieces of syntax might be OT, but their external affairs appear to be Clash & Crash.

Likewise, compare what happens when movement of an adverb crosses an island — a case considered in (5.26) — with what happens when movement of an NP crosses the same sort of island — a case considered in (5.68). Both examples are repeated below.[31]

(5.76) *I'm wondering how the company will fire [any employees who treat their subordinates __].

(5.77) **a.** *There is one worker who the company actually fired [the employee that treated __ badly].
b. There is one worker who the company actually fired [the employee that treated him badly].

Island constraints prohibit a gap. Presumably the same island constraints are responsible for the ungrammaticality of both (5.76) and (5.77a). For (5.77a), in the true spirit of OT, there is a repair strategy — a minimal violation of SILENT TRACE (the resumptive pronoun) that permits the higher–ranked island constraint to be satisfied. In (5.76), however, there is no successful repair strategy. No resumptive pronoun for the adverb can save (5.76). Not even *thus* will work, in dialects that use this word as an adverbial pronoun:[32]

(5.78) *I'm wondering *how* the company will fire any employees who treat their subordinates *thus*.

On the face of it, it looks as if any possibility that the pronunciation system offers for resumptive adverbs is ruled out by a constraint completely outside the pronunciation system. The constraint must mark as deviant the whole class of outputs from that system.

(5.79) **NO RESUMPTIVE ADVERBS:** There are no resumptive adverbs.
[Interacts with the pronunciation system in Clash & Crash fashion.]

Now an OT purist might at this point attempt to *reformulate* the Case Filter, principles governing adverbs, notions of faithfulness, and other aspects of the proposal, so as to cast an OT light on the entire set of problems. Perhaps the attempt would be successful in the areas discussed above. But there is one more category of evidence which suggests that not all is OT.

Consider the consequences of one simple and obvious fact about resumptive pronouns in English and French. True, a resumptive pronoun "saves" a structure that would otherwise violate an island constraint — but the structure with the resumptive pronoun is only marginally acceptable. (This fact is reflected in the restriction of such pronouns to casual registers of speech.) Gradient judgments fall outside of OT as a simple matter of logic. The system — as it is set up, and as it functions correctly in many domains — picks

[31]To make the cases more comparable, we really should use a relative clause instead of the question in (5.26), but it makes no difference to the outcome:
(i) *This is the way the company will fire [any employees who treat their subordinates __].

[32]Example (5.78) is fine, of course, with an irrelevant interpretation in which *how* talks about methods of *firing* (not methods of treating subordinates) and in which *thus* is not a resumptive pronoun for *how*, but a demonstrative pronoun with some other antecedent.

winners and losers. There are no silver or bronze medals in OT. Consequently, when we find silver and bronze medalists among possible pronunciations for certain structures, we can be certain that there are non–OT aspects to the interaction of linguistic constraints. From that point on, we *know* that there is a division of labor to be discovered. This chapter has made a preliminary attempt to discern what this division of labor might be.

Before concluding this chapter, it has to be noted that every aspect of this discussion is controversial. It is simply impossible in 1996 to identify a successful mainstream of work on OT interactions within syntax. Too little has been accomplished and too much is unknown. Grimshaw (to appear), for example, analyzes a variety of phenomena governing movement and structure in OT terms. Anderson (to appear) offers an approach to verb–second phenomena in a similar vein. Speas (this volume) does the same in the domain of anaphora — though I suspect that the phenomena she discusses may fall under the pronunciation system discussed here. In each case, a serious comparison of these suggestions with Clash & Crash alternatives still lies in the future. Nonetheless, even among the topics discussed in this chapter, there are some hints that some constraints on structure and movement may have an OT character after all.

For example, many non–literary registers of French allow apparent violations of our system — questions and relative clauses in which both the *wh*–phrase and the complementizer are pronounced.

(5.80) Pronunciation of *wh*– and C in non–literary French
Je me demande *quand que* Pierre arrivera [non–literary]
I wonder when that Pierre will arrive

There is some reason to think that these structures are not the counterexamples that they seem. In particular, it can be argued that these structures actually involve two distinct CPs — one containing the *wh*–phrase and an unpronounced complementizer, and another containing the pronounced complementizer. If (5.81) is the structure of (5.80), then each CP satisfies the pronunciation constraints of French. In the higher CP, RECOVERABILITY requires LEFTEDGE(CP) to be violated, so lower–ranked TELEGRAPH gets a chance to be satisfied. In the lower CP, LEFTEDGE(CP) is satisfied, though lower–ranked TELEGRAPH is violated.

(5.81) Je me demande [$_{\text{CP}}$ *quand* ~~*que*~~ [$_{\text{CP}}$ *que* Pierre arrivera]]
I wonder when ~~that~~ that Pierre will arrive

I will not present the arguments for (5.81) here. (They involve complexities in the syntax of *quoi* 'what' that would require a lengthy excursus.) My point is different. If (5.81) is correct, then we have to wonder why double CPs are *only* found when otherwise no clause would satisfy LEFTEDGE(CP). For example, while complex relative clauses allow the structure seen in (5.81), simple relative clauses do not.

(5.82) l'homme [$_{\text{CP}}$ avec qui [$_{\text{CP}}$ que j'ai dansé]] [non–literary French]
the man with whom that I danced
structure: l'homme [$_{\text{CP}}$ avec qui ~~que~~ [$_{\text{CP}}$ que j'ai dansé]]

(5.83) *l'homme [$_{CP}$ que [$_{CP}$ que je connais]]
the man that that I know
structure: l'homme [$_{CP}$ ~~qui~~ que [$_{CP}$ que j'ai dansé]]

If the proposed structure is correct in the first place, it looks as though the double CP construction is only possible when otherwise LEFTEDGE(CP) would not be satisfied in the lower of the two CPs. (I thank Denis Bouchard for this observation.) When LEFTEDGE(CP) can be satisfied in this CP without the recursion, the multiple CP structure is impossible. This observation smells enough like OT to make one wonder whether this phenomenon will not in the end support an extension of OT interactions beyond the domain of sentence pronunciation. Furthermore, in the domain of movement, it is often suggested that movement is avoided whenever it is not necessary in order to satisfy some constraint. Such **economy** conditions also smell like OT (a point also noted in Chapter 6), and have been incorporated into the systems of Grimshaw and others. Could it be the case that the internal workings of movement and structure–building also display OT organization — but perhaps interact with a separate pronunciation system in a Clash & Crash fashion? To these and many other questions, we do not yet know the answers.